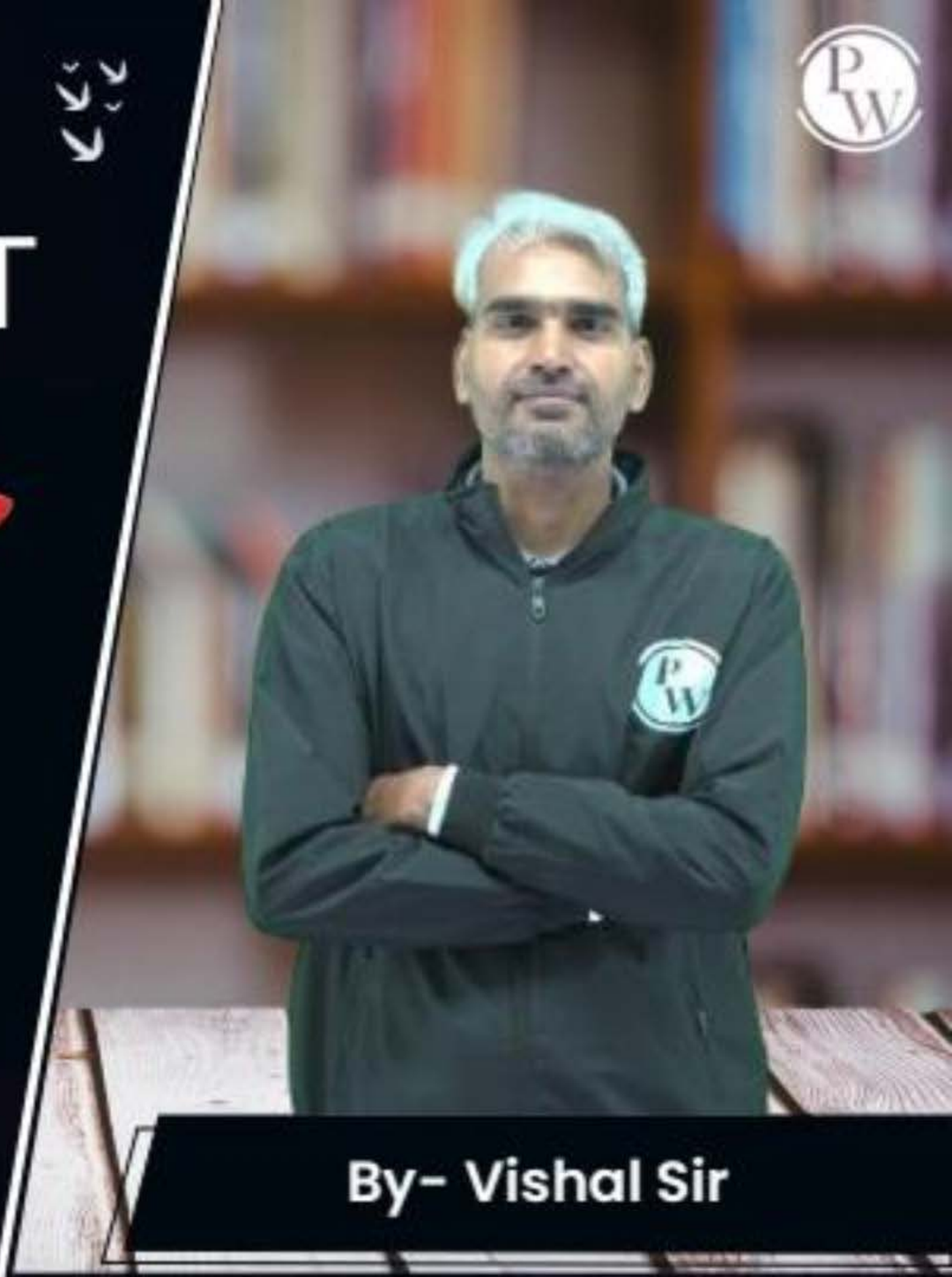
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Classification based on serializability
- **Topic** Conflicting and non-conflicting operations
- **Topic** Conflict equivalent schedule
- **Topic** Conflict serializable schedule
- **Topic** Precedence graph
- **Topic** Topological order

R(A) - W(A)
W(A) - R(A)
W(A) - W(A)

R(A) — R(A)
R(A)/W(A) — R(B)/W(B)

Slide

Topics to be Covered



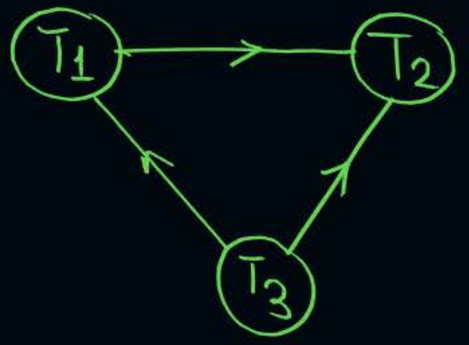
- **Topic** Classification based on serializability
- **Topic** View serializable schedule
- **Topic** View equivalence condition

Slide

#Q. Check whether the schedule is conflict serializable schedule or not?

If conflict serializable schedule then identify all conflict equivalent serial schedules.

T_1	T_2	T_3
		$R_3(Y)$
		$R_3(Z)$
$R_1(X)$		
$W_1(X)$		
		$W_3(Y)$
		$W_3(Z)$
	$R_2(Z)$	
$R_1(Y)$		
$W_1(Y)$		
	$R_2(Y)$	
	$W_2(Y)$	
	$R_2(X)$	
	$W_2(X)$	



Acyclic precedence graph

∴ Conflict Serializable Schedule

Topological order:-

$T_3 \longrightarrow T_1 \longrightarrow T_2$

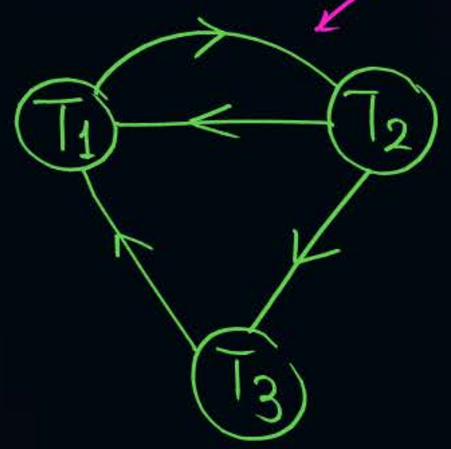
∴ given schedule is Conflict equivalent to Serial Schedule

$T_3 \longrightarrow T_1 \longrightarrow T_2$

#Q. Check whether the schedule is conflict serializable schedule or not?

If conflict serializable schedule then identify all conflict equivalent serial schedules.

T_1	T_2	T_3
	$R_2(A)$	
$R_1(B)$		
	$W_2(A)$	
		$R_3(A)$
$W_1(B)$		
$W_1(A)$		
	$R_2(B)$	
	$W_2(B)$	

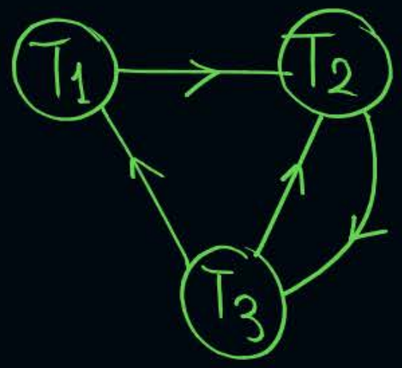


Cycle
∴ Not a Conflict serializable schedule.

#Q. Check whether the schedule is conflict serializable schedule or not?

If conflict serializable schedule then identify all conflict equivalent serial schedules.

T_1	T_2	T_3
		$W_3(B)$
$R_1(A)$		
$W_1(B)$		
	$R_2(B)$	
	$W_2(C)$	
		$R_3(C)$



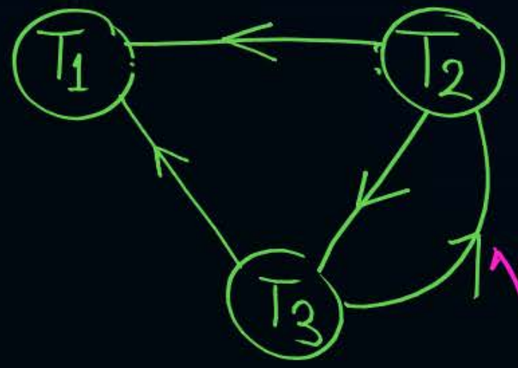
Cycle

∴ Not a Conflict serializable schedule

#Q. Check whether the schedule is conflict serializable schedule or not?

If conflict serializable schedule then identify all conflict equivalent serial schedules.

T_1	T_2	T_3
	$R_2(z)$	
	$R_1(y)$	
	$W_2(y)$	
		$R_3(y)$
		$R_3(z)$
$R_1(x)$		
$W_1(x)$		
		$W_3(y)$
		$W_3(z)$
	$R_2(z)$	
$R_1(y)$		
$W_1(y)$		

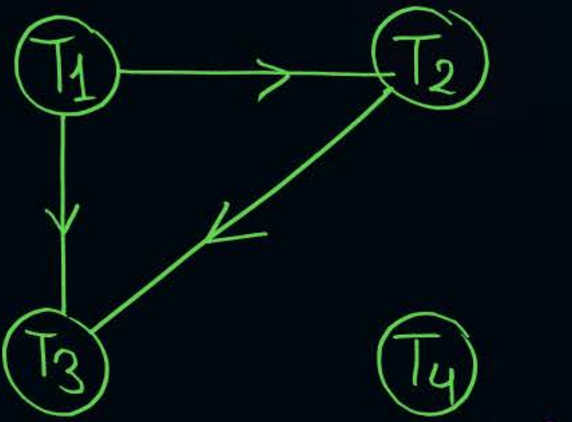


Cycle
∴ Not a Conflict serializable Schedule

#Q. Check whether the schedule is conflict serializable schedule or not?

If conflict serializable schedule then identify all conflict equivalent serial schedules.

T_1	T_2	T_3	T_4
$R_1(A)$			
	$R_2(A)$		
		$R_3(A)$	
			$R_4(A)$
$W_1(B)$			
	$W_2(B)$		
		$W_3(B)$	



Acyclic precedence graph

∴ Schedule is Conflict Serializable Schedule.

Topological orders:

$T_4 \rightarrow T_1 \rightarrow T_2 \rightarrow T_3$

$T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$

$T_1 \rightarrow T_4 \rightarrow T_2 \rightarrow T_3$

Given Schedule is Conflict equivalent to serial schedules

$T_4 \rightarrow T_1 \rightarrow T_2 \rightarrow T_3$, $T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4$

$T_1 \rightarrow T_2 \rightarrow T_4 \rightarrow T_3$ & $T_1 \rightarrow T_4 \rightarrow T_2 \rightarrow T_3$

#Q. Check whether the schedule is conflict serializable schedule or not?

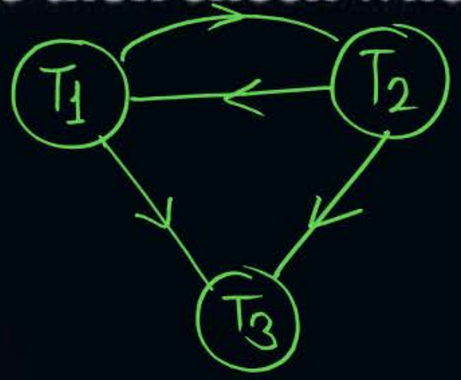
If not conflict serializable then check where the schedule is serializable schedule or not?

S

T_1	T_2	T_3
$R_1(A)$		
	$W_2(A)$	
$W_1(A)$		
		$W_3(A)$

Read from initial DB

final writer of A is T_3



Cyclic precedence graph
∴ Not a Conflict serializable schedule

Serial schedule
$T_1 \rightarrow T_2 \rightarrow T_3$

T_1	T_2	T_3
$R_1(A)$		
$W_1(A)$		
	$W_2(A)$	
		$W_3(A)$

Read from initial DB

final writer of A is T_3

We can observe
$S \equiv T_1 \rightarrow T_2 \rightarrow T_3$

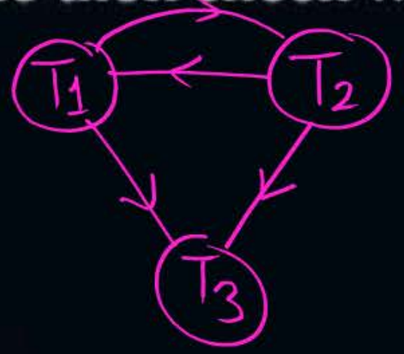
Schedule 'S' is not a Conflict serializable schedule but it is a serializable schedule

#Q. Check whether the schedule is conflict serializable schedule or not?

If not conflict serializable then check where the schedule is serializable schedule or not?

S_2

T_1	T_2	T_3
	$W_2(A)$	
$W_1(A)$		
	$W_2(A)$	
		$W_3(A)$



Cyclic precedence graph

∴ Not a Conflict serializable schedule

Schedule S_2 is equivalent to serial schedules

$T_1 \rightarrow T_2 \rightarrow T_3$

& $T_2 \rightarrow T_1 \rightarrow T_3$

Schedule S_2 is Not a Conflict serializable schedule, but a Serializable schedule

Note°- Conflict serializability Condⁿ is "if-then" Condition.

i.e. If schedule is conflict serializable schedule then schedule is serializable schedule.

& If "If" Condⁿ is false then we can not say anything about "then" statement

i.e, If schedule is not a conflict serializable schedule then schedule may or may not be serializable.

Note:- View serializability Condition is "if and only if" Condⁿ.

i.e. Schedule is View serializable schedule
if and only if
Schedule is a serializable schedule.



Topic : View serializable schedule

If a given schedule is view equivalent to at least one of the serial schedule , then it is called a view serializable schedule.



Topic : View equivalent condition

Consider two schedules S1 and S2

Schedules S1 and S2 are called view equivalent if the following three conditions hold true for them –

- Condition-01 : For each data item X, if transaction Ti reads X from the initial database in schedule S1, then in schedule S2 also, Ti must perform the initial read of X from the initial database.

Thumb Rule

- "Initial readers must be same for all the data items".



Topic : View equivalent condition

Condition-02 :

If transaction Ti reads the value of data item X updated by the transaction Tj in schedule S1, then in schedule S2 also, transaction Ti must read the value of data item X updated by the transaction Tj.

Thumb Rule

- "Write-read sequence must be same.".



Topic : View equivalent condition

Condition-03 :

For each data item X, if X is finally updated by transaction Ti in schedule S1, then in schedule S2 also, X must be finally updated by transaction Ti.

Thumb Rule

- "Final writers must be same for all the data items".

* All three conditions must hold true for two schedules to be view equivalent.

Q: Check whether the schedule is a view serializable schedule or not? If view serializable schedule then identify all view equivalent serial schedules.

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_2(B)$		$W_2(B)$

① Read from initial DB Condition

Data item	Transaction those are reading the value of data item from initial DB	Other transaction those are updating that data item	Constraints wrt equivalent serial schedule
A	No transaction	T_1	(No Constraint)
B	T_2	T_1, T_3	$T_2 \rightarrow (T_1, T_3)$ $\equiv \{T_2 \rightarrow T_1, T_2 \rightarrow T_3\}$

Q: Check whether the schedule is a view serializable schedule or not? If view serializable schedule then identify all view equivalent serial schedules.

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_2(B)$		
		$W_2(B)$

(2) Write-Read Sequence Condition

In the given schedule if Transaction Tj Reads the value of data item A updated by transaction Ti,

then in equivalent serial schedule

Ti must execute before Tj

i.e $T_i \longrightarrow T_j$

In between Ti & Tj no other writer of data item A should be allowed to execute

Q: Check whether the schedule is a view serializable schedule or not? If view serializable schedule then identify all view equivalent serial schedules.

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_2(B)$		
		$W_2(B)$

(2) Write-Read Sequence Condition

data item A: In given schedule T_2 Reads the value of A updated by T_1.

∴ In equivalent serial schedule $T_1 \rightarrow T_2$

Anyway there is no other write of data item 'A'

Constraint because of write Read seq of data item A

data item B:

No write-read sequence w.r.t data item B ∴ No Constraint wrt data item B.

Q: Check whether the schedule is a view serializable schedule or not? If view serializable schedule then identify all view equivalent serial schedules.

T_1	T_2	T_3
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_2(B)$		
		$W_2(B)$

③ Final update Condition

data item	Transaction with final update of that data item	other writers of that data item	Constraints w.r.t equivalent serial schedule
A	T_1	No other transaction	No Constraint
B	T_3	(T_1, T_2)	$(T_1, T_2) \longrightarrow T_3$ $\equiv \{T_1 \longrightarrow T_3,\ T_2 \longrightarrow T_3\}$

Note: Once we have obtained the Constraints based on all three Conditions, then Construct the (Precedence) dependency graph based on overall constraints.

If acyclic precedence graph then view serializable schedule and hence a serializable schedule

Overall Constraints w.r.t. above example

Wrt Read from initial DB Condition → $T_2 \rightarrow T_1$, $T_2 \rightarrow T_3$

Wrt. write-read sequence Condn → $T_1 \rightarrow T_2$

Wrt final update Condn → $T_1 \rightarrow T_3$, $T_2 \rightarrow T_3$

Cycle ∴ Not a View Serializable Schedule

In overall Constraints if $T_i \rightarrow T_j$ & $T_j \rightarrow T_i$ then it corresponds to cycle in the precedence graph ∴ Schedule will not be View Serializable Schedule

Q: Check whether the schedule is a view serializable schedule or not? If view serializable schedule then identify all view equivalent serial schedules.

T_1	T_2	T_3
		$R_3(B)$
$W_1(A)$		
	$R_2(B)$	
	$R_2(A)$	
	$W_2(B)$	
$W_2(B)$		
		$W_2(B)$

① Read from initial DB Condition

data item	Transaction Reading from initial DB	other writers of Data item	Constraints
A	NO	T_1	NO Constraint
B	T_2	(T_1, T_3)	$T_2 \to (T_1, T_3) \equiv T_2 \to T_1$, $T_2 \to T_3$
	T_3	(T_1, T_2)	$T_3 \to (T_1, T_2) \equiv T_3 \to T_1$, $T_3 \to T_2$ (Not V.S.S.)

② Write Sequence Cond^n:
Same as previous question

③ Final update Cond^n:-
Same as previous question

Q: Check whether the schedule is V.S.S or not? If V.S.S then identify all view equivalent serial schedules

(S)

T_1	T_2
W(A)	
	R(A)
	W(B)
R(B)	

① Read from initial DB Cond^n

A: No Constraint.

B: No Constraint.

② Write Read Seq Cond^n

A: Constraint is $T_1 \longrightarrow T_2$ { No other write of A }

B: Constraint is $T_2 \longrightarrow T_1$ { Not other write of B }

∴ Cycle

Hence Not a V.S.S.

& hence Not a serializable schedule

③ Final update Cond^n

A: No Constraint

B: No Constraint

Q: Check whether the schedule is V.S.S or not? If V.S.S then identify all view equivalent serial schedules

T_1	T_2	T_3
	R(B)	
	R(A)	
W(B)		
W(A)		
	W(A)	
		W(A)

① Read from initial DB Cond^n

A: Constraint is $T_2 \rightarrow (T_1, T_3)$

B: Constraint is $T_2 \rightarrow T_1$

② Write Read Seq Cond^n

A: No WR Seq $\therefore$ No Constraint

B: No WR Seq $\therefore$ No Constraint

③ Final update Cond^n

A: Constraint is $(T_1, T_2) \longrightarrow T_3$

B: No Constraint

Overall Constraints

$T_2 \rightarrow T_1$

$T_2 \rightarrow T_3$

$T_1 \rightarrow T_3$

$\Downarrow$

Precedence graph w.r.t. Constraints.



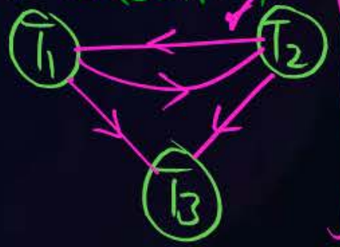
Acyclic

$\therefore$ Schedule is V.S.S

&

View equivalent serial schedule is $T_2 \rightarrow T_1 \rightarrow T_3$

Precedence graph w.r.t Conflict Serializability



Cycle

$\therefore$ Not Conflict Serializable Schedule



2 mins Summary

Topic Classification based on serializability

Topic View serializable schedule

Topic View equivalence condition

Slide

THANK - YOU